The **brain** is the part of the body which lets living beings think. It does some bodily functions, such as telling the rest of the body what to do. Almost all animals have a brain: the exceptions are sponges, cnidarians, and lancelets. Plants and fungi do not have brains, although they do react to changes in their environment.

The brain is made up of special cells called nerves, which are connected with each other and with other nerves in our body. The brain gets input from sense organs, and changes behavior in response to this information. In humans, the brain also controls our use of language, and is capable of abstract thought.[1] The brain is the main control centre of the whole body.[2]

In all animals, the brain is protected in some way. In ourselves, and all vertebrates, it is protected by the bones of the skull.

Function

The brain does the thinking, learning, and feeling for the body. For humans, it is the source of consciousness.[3] The brain also controls basic autonomic body actions, like breathing, digestion, and heartbeat, that happen automatically. These activities, and much else, are governed by unconscious functions of the brain and nervous system. All the information about the world gathered by our senses is sent through nerves into the brain, allowing us to see, hear, smell, taste, and feel things. The brain processes this information, and we experience it as pictures, sounds, and so on. The brain also uses nerves to tell the body what to do, for example by telling muscles to move or our heart to beat faster.

This is generally true but some activity is caused by the spinal cord directly. For example, reflex actions do not involve the brain. In lower animals, a lot is done without their brain being involved.

All vertebrates have brains and, over time, their brains have evolved to become more complex.[4] Some simple animals, however, like sponges, do not have anything like a brain. Segmented invertebrates have ganglions in each segment, and a ring of nervous tissue around the alimentary canal at the front. This acts to bring sense data from the front into play with the movement of the body.

Parts

In mammals, the brain is made of three main parts: the cerebrum, the cerebellum, and the brainstem. The surface of the cerebrum is the cerebral cortex, which all vertebrates have. Mammals also have an extra layer, the neocortex. This is the key to the behaviour which is typical of mammals, especially humans.[5]

Cerebral cortex

The cortex has sensory, motor, and association areas. The sensory areas are the areas that get and process information from the senses. The motor areas control voluntary movements, especially fine movements performed by the hand. The right half of the motor area controls the left side of the body, and vice versa. Association areas produce a meaningful experience of the world and supports abstract thinking and language. This enables us to interact effectively. Most connections are from one area of the cortex to another, rather than to subcortical areas; The figure may be as high as 99%.[6]

Thalamus

The thalamus puts together the input from most of the senses. It is the place in the brain which makes the "picture" we have of the world outside us. The thalamus sits centrally under the cerebral cortex.

Cerebellum

The cerebellum coordinates muscles so they work together.[7] It is also the centre of keeping position and balance, and motor skills. It is one of the most ancient parts of the brain and sits at the back underneath the cerebral cortex.

Brain stem

The brain stem is at the back of the brain (actually underneath it in humans). It joins the rest of the brain with the spinal cord. It has lots of different parts that control different jobs in the body: for instance, the brain stem controls breathing, heartbeat, sneezing, eye blinking, and swallowing. Body temperature and hunger are also controlled by parts of the brain stem. Roughly speaking, it deals with the more automated aspects of nervous function.

Size

The volume of the human brain (relative to the size of the whole body) is very large, compared to that of most other animals. The human brain also has a very large surface (called cortex) for its size, which is possible because it is very wrinkled. If the human cortex were flattened, it would be close to a square meter in area. Some other animals also have very wrinkled brains, such as dolphins and elephants. Here is a rule of thumb: the larger an animal is, the larger its brain will be.[1]p15 Even allowing for that, the human brain, and in particular the neocortex, is very large. We know it increased in size *four-fold* over the last several million years of evolution.[1]p79 There are ideas about why this happened, but no-one is quite sure. Most theories suggest complex social activity and the evolution of language would make a larger brain advantageous.[1]p80[8] As an additional note, Einstein's brain weighed only 1,230 grams, which is less than the average adult male brain (about 1,400 grams).[9] The detailed organisation of a brain obviously matters, but in ways which are not understood at present.

Number of cells

A human brain accounts for about 2% of the body's weight, but it uses about 20% of its energy. It has about 50–100 billion nerve cells (also called neurons), and roughly the same number of support cells, called *glia*. The job of neurons is to receive and send information to and from the rest of the body, while glia provide nutrients and guide blood flow to the neurons, allowing them to do their job. Each nerve cell has contact with as many as 10,000 other nerve cells through connections called synapses.

Related pages

Neural signaling in the human brain. Small electrical charges go from one neuron to the next

The **thalamus** (plural: **thalami**; from Greek: θάλαμος, trans. "inner chamber")[1] is a midline symmetrical structure in the brains of vertebrates. The thalamus is situated between the midbrain and cerebral cortex.

Overview

The thalamus relays sensory and motor signals to the cerebral cortex,[2][3] and regulates consciousness, sleep and alertness.

The thalamus sits above the hypothalamus but below the cerebral cortex. It is a collection of nuclei with various functions. It acts as a relay station, gathering sensory information – except olfactory – and passes it on to the cerebral cortex.

There are action systems[*clarification needed*] for various types of behaviour, including but not limited to eating, drinking, defecation and copulation.[4] These behaviours satisfy short-term needs, and are called 'consummatory' behaviours.

Functions

The thalamus has many functions. First of all, it acts as a relay station, or hub. It relays information between subcortical areas and the cerebral cortex.[5] In particular, every sensory system except smelling has a thalamic nucleus that receives sensory signals and sends them to the related areas in the cortex.

Examples

For sight, inputs from the retina are sent to the thalamus, which in turn sends them to the visual cortex in the occipital lobe.

Sensory processing

The thalamus is believed to process sensory information as well as to relay it: each of the main sensory relay areas gets strong feedback from the cerebral cortex.[6]

Sleep regulation

The thalamus also plays an important role in regulating states of sleep and wakefulness.[7] Thalamic nuclei have strong connections with the cerebral cortex. These circuits are believed to be involved with consciousness.[8] The thalamus plays a major role in regulating arousal, the level of awareness, and activity. Damage to the thalamus can lead to permanent coma.[9]

Basal ganglia

The thalamus has a role in the basal ganglia system, but the mechanism is poorly understood. The thalamus has been thought of as a "relay" that just forwards signals to the cerebral cortex. But research suggests that thalamus is more selective.[10]

Sensory systems

Many functions are linked to regions of the thalamus. This is the case for most sensory systems, except the olfactory system, such as the auditory, somatic sensory system, visceral, eating and visual systems. There are specific lesions cause specific sensory deficits. A big role of the thalamus is the support for motor and language systems. Much of the circuitry for these systems is shared with the thalamus. The thalamus is *functionally connected* to the hippocampus.[11] This is part of the hippocampal system which is crucial for human episodic event memory.[12][13][14]

Motor control

The information for motor control is a network involving the thalamus as a subcortical motor center.[15] In the brains of primates, the thalamus provides the specific channels from the basal ganglia and cerebellum to the cortical motor areas.[16][17][18]

Research

In an investigation of the eye movement motor response in three monkeys, the thalamic regions were found to cause antisaccade eye movement. That is the ability to inhibit the reflexive jerking movement of the eyes in the direction of a presented stimulus.[19][20] They still look in the direction of the stimulus, but do so in a more controlled manner.

Recent findings

Recent findings suggested that the mediodorsal (MD) thalamus may "amplify the connectivity (signaling strength) of just the circuits in the cortex needed for the current context, aiding in the flexibility – of the mammalian brain – to make complex decisions by wiring the many associations on which decisions depend into weakly connected cortical circuits."[21]

Researchers also found that "enhancing MD activity magnified the ability of mice to think".[21] This lowered by more than 25 percent their error rate in deciding which conflicting sensory stimuli to follow to find the reward."[22]

Summary

In short, the thalamus helps to make mammals more effective at making decisions and living in their natural environment.

The half-second delay

The thalamus is a key area which is involved in what is known as the "half-second delay". This is a perceptual illusion which has only recently been discovered.[23][24] The discovery is as follows: we perceive events in the world as instantaneous. They happen when we see (hear, sense) them happening. But the processing of visual signals at least is what the thalamus does.

Experientially, we see events happening (so we think) without any delay. But in reality *it takes half a second for the brain to organise the data* which comes in from, let us say, vision. However, we do not sense that half-second delay at all.

Our perception of events is that we see them as they happen,[25] which applies to consciousness and conscious behaviour. Automatic responses to sudden pain happen instantaneously (bare foot on a pin for example), but they are actually perceived a moment after.